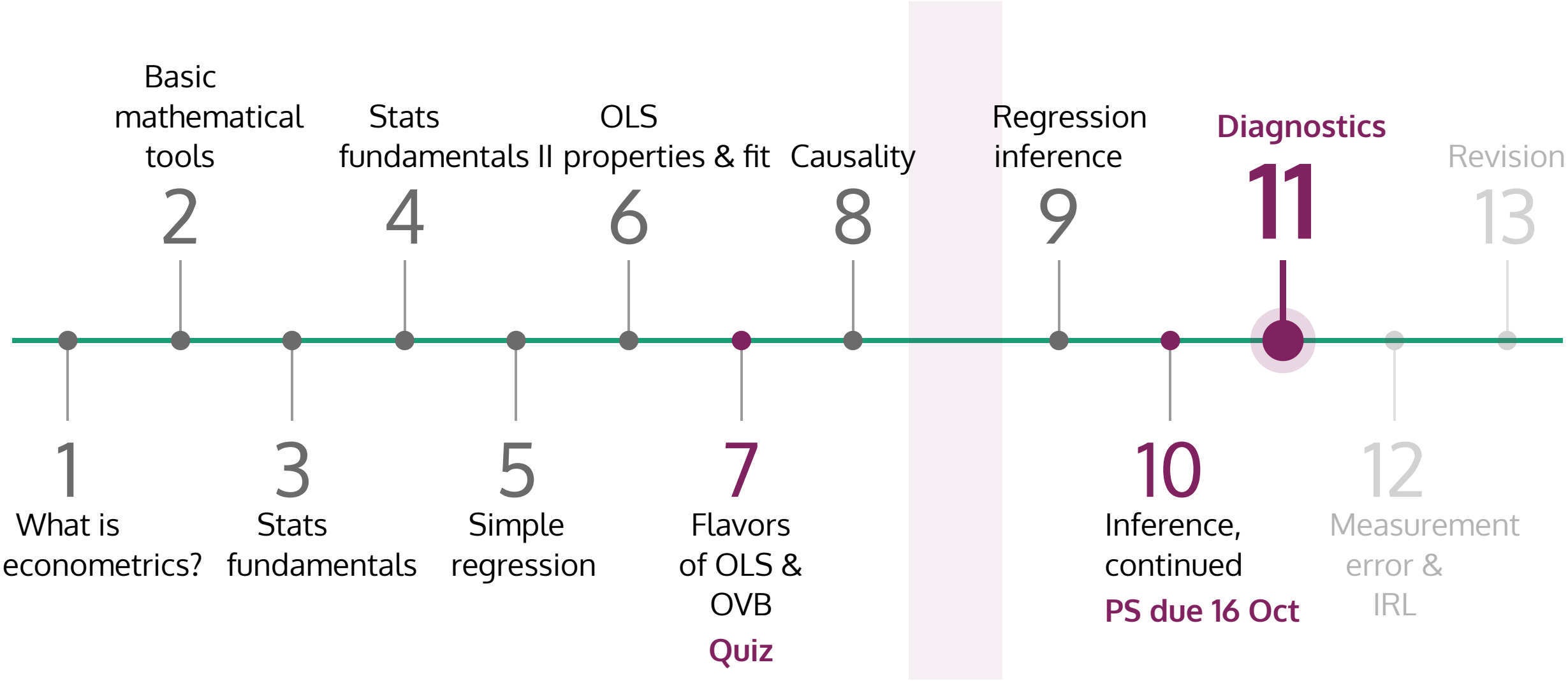

ECON 2041
Introductory Econometrics
Testing assumptions

Plan for the semester



Plan for today

- Residual plots: a portal game
- Functional form: happiness and GDP
- Heteroskedasticity: Airbnb prices
- Robust standard errors

Guess the residual plot

A residual is actual Y minus fitted Y

$$\hat{u}_i = y_i - \hat{y}_i$$

Plot the residuals against the fitted values $\hat{y}_i$

What we want to see: pure noise

Open the portal on your phone

On the next slide: 4 residual plots

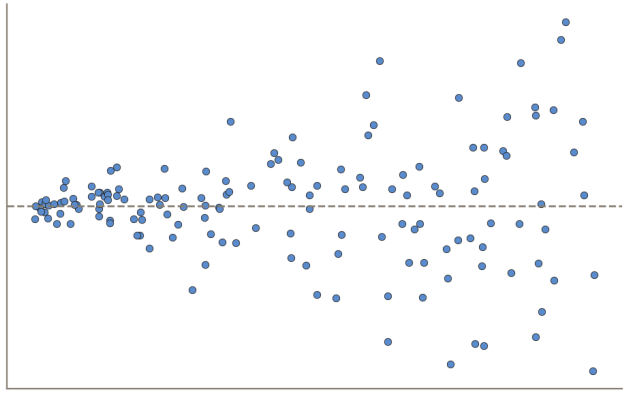
Tick every plot whose residuals look like pure noise

https://emiliatjernstrom.com/econ2041/portal.html

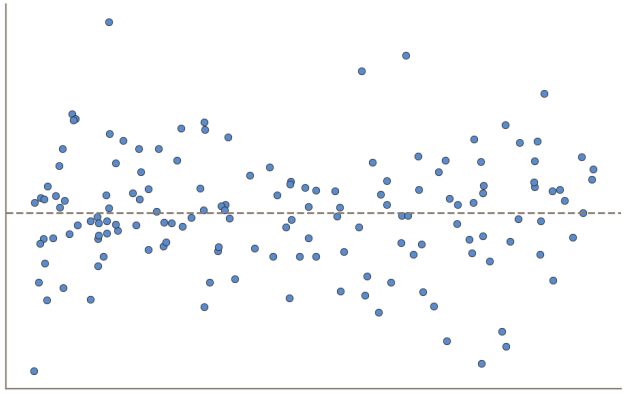


In which plots are the residuals pure noise?

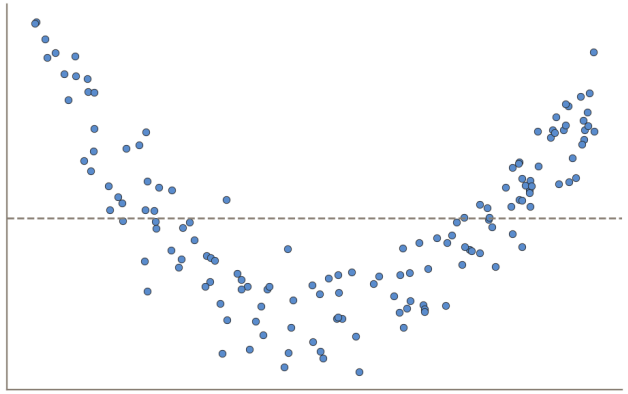
A: unequal residual variance



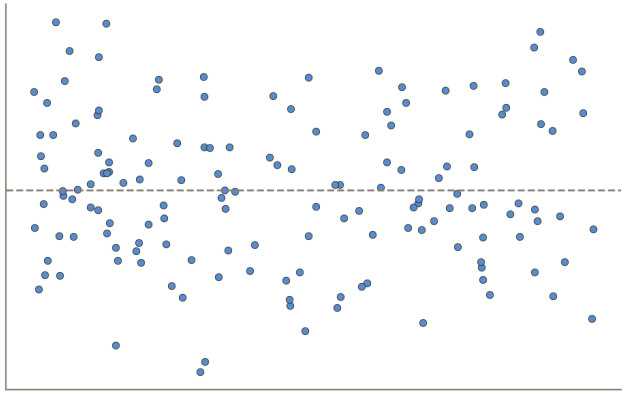
B: pure noise



C: a curved pattern



D: pure noise

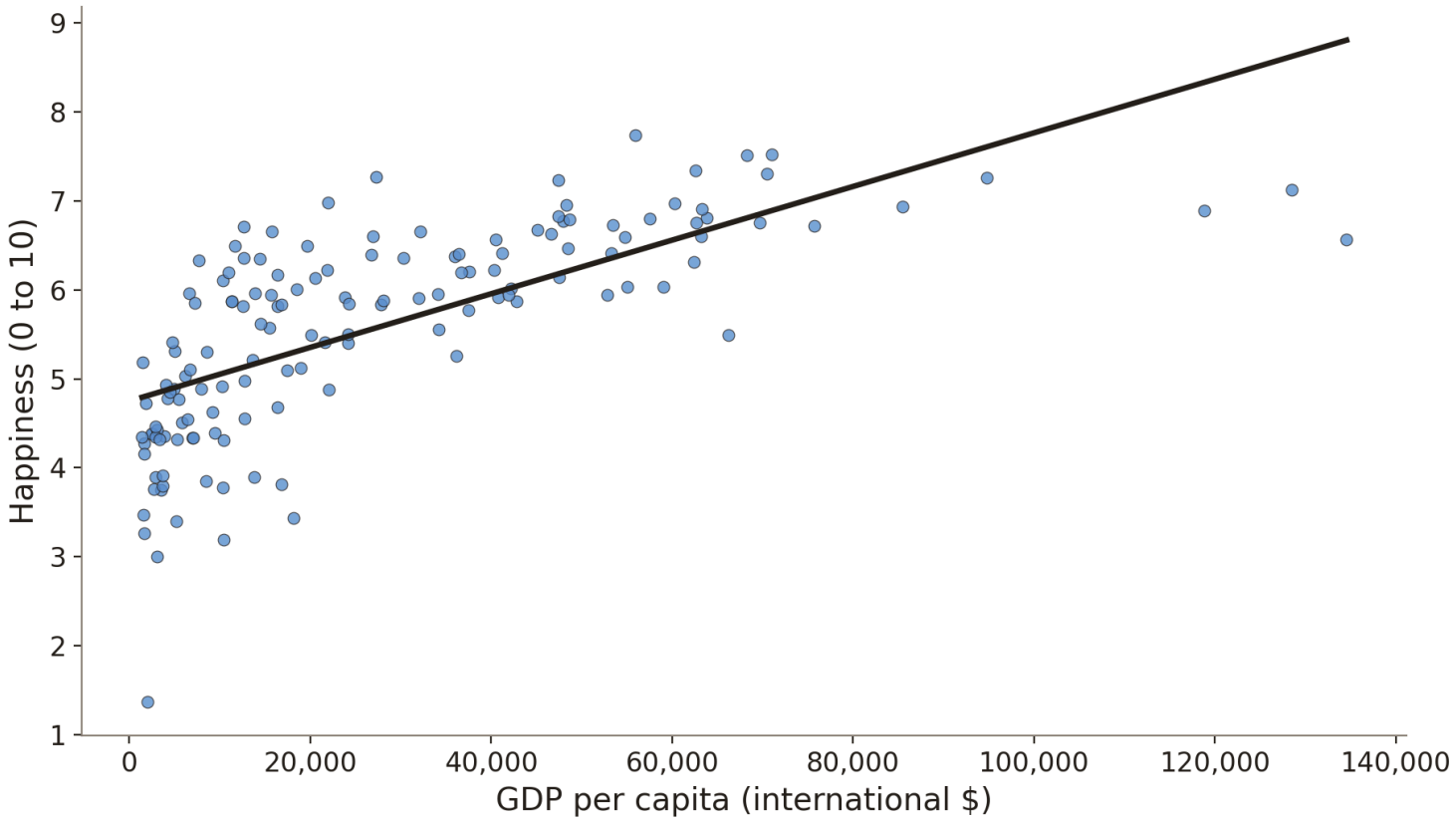


Vote on the portal: https://emiliatjernstrom.com/econ2041/portal.html

Is happiness linear in GDP?

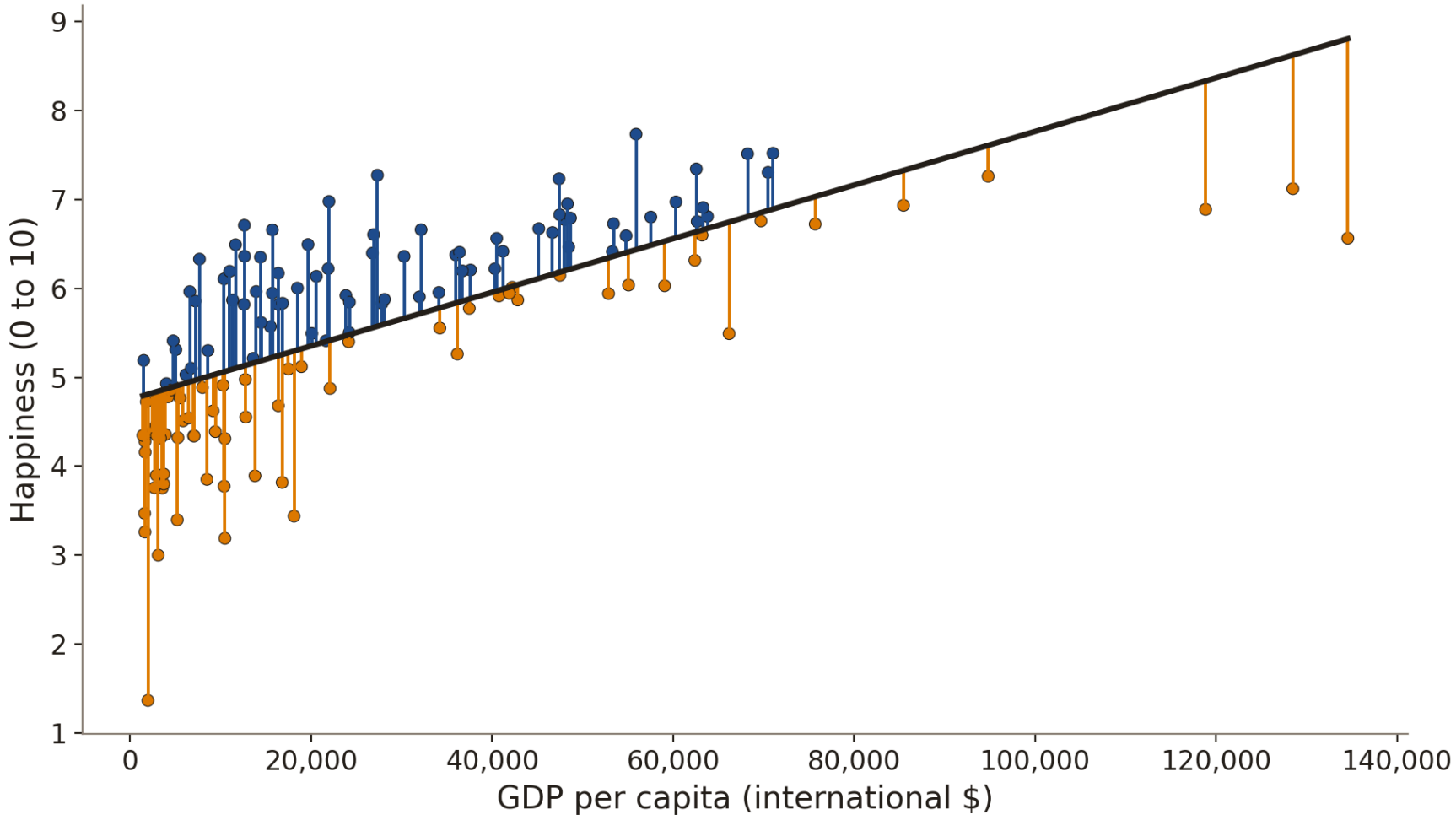
Richer countries are happier

$$\widehat{\text{happiness}} = 4.75 + 0.03 \times \text{GDP per capita (\$1,000s)}$$



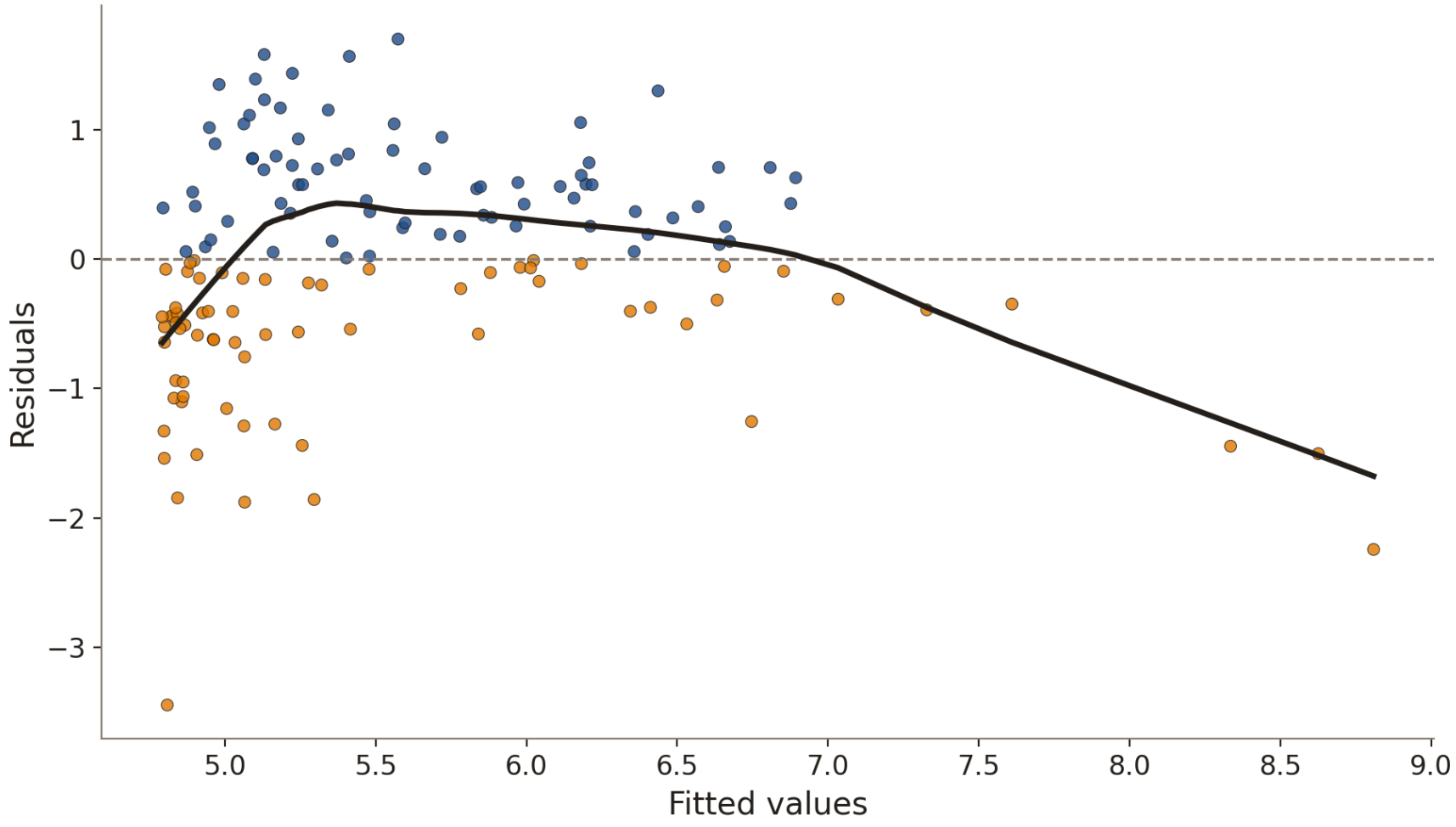
Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

Residuals: negative, positive, then negative



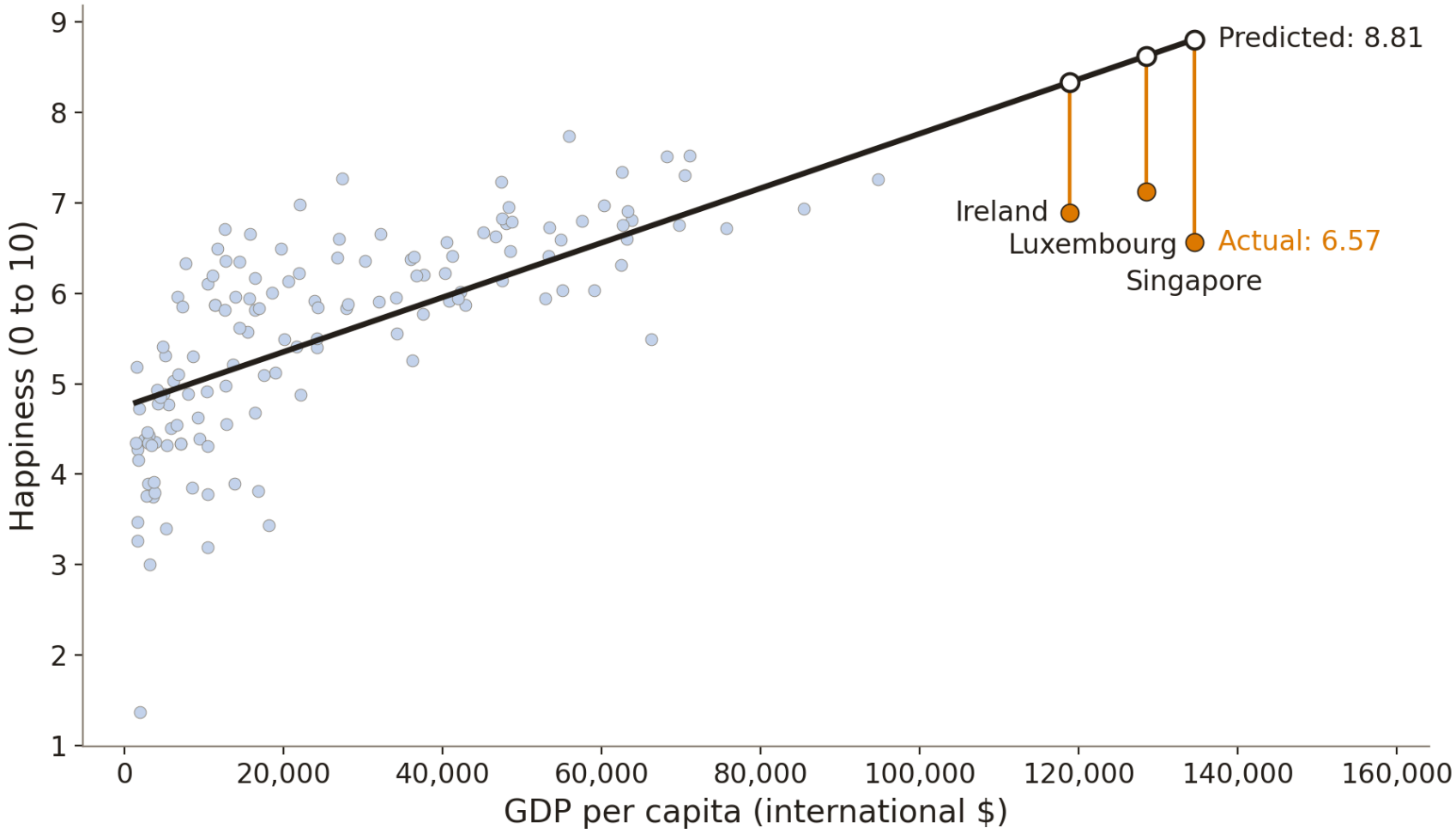
Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

A curved pattern in the residual plot



Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

Richest countries: below their fitted values



Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

We reject linearity with the RESET test

```
from statsmodels.stats.diagnostic import linear_reset

model = ols("happiness ~ gdp_k", data=happy).fit()
linear_reset(model, power=3, use_f=True)
```

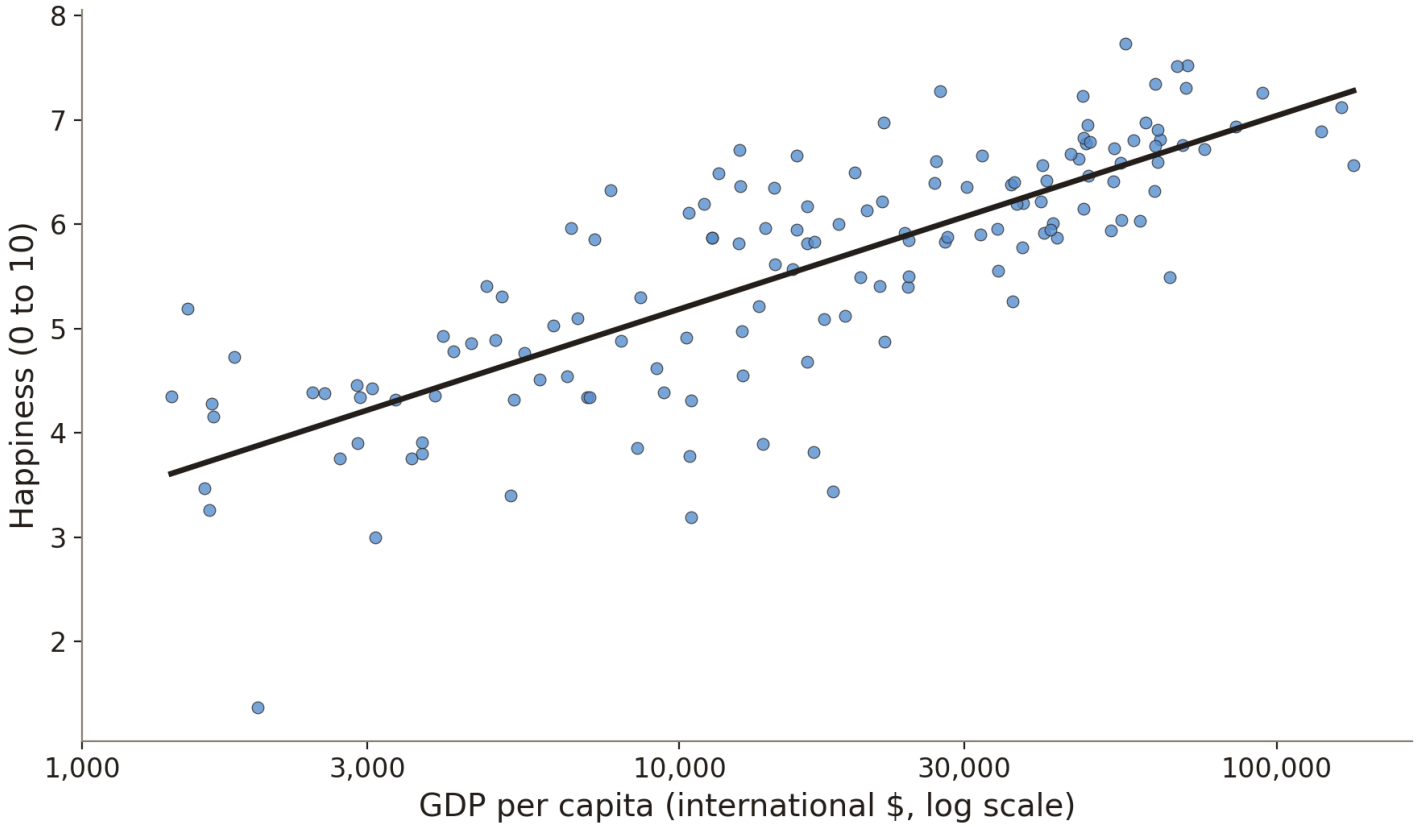
```
<F test: F=25.21835447783917, p=4.430122822451849e-10, df_denom=140, df_num=2>
```

p is below 0.1, so we reject the null that the coefficients on the squared and cubed fitted values are both 0

Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

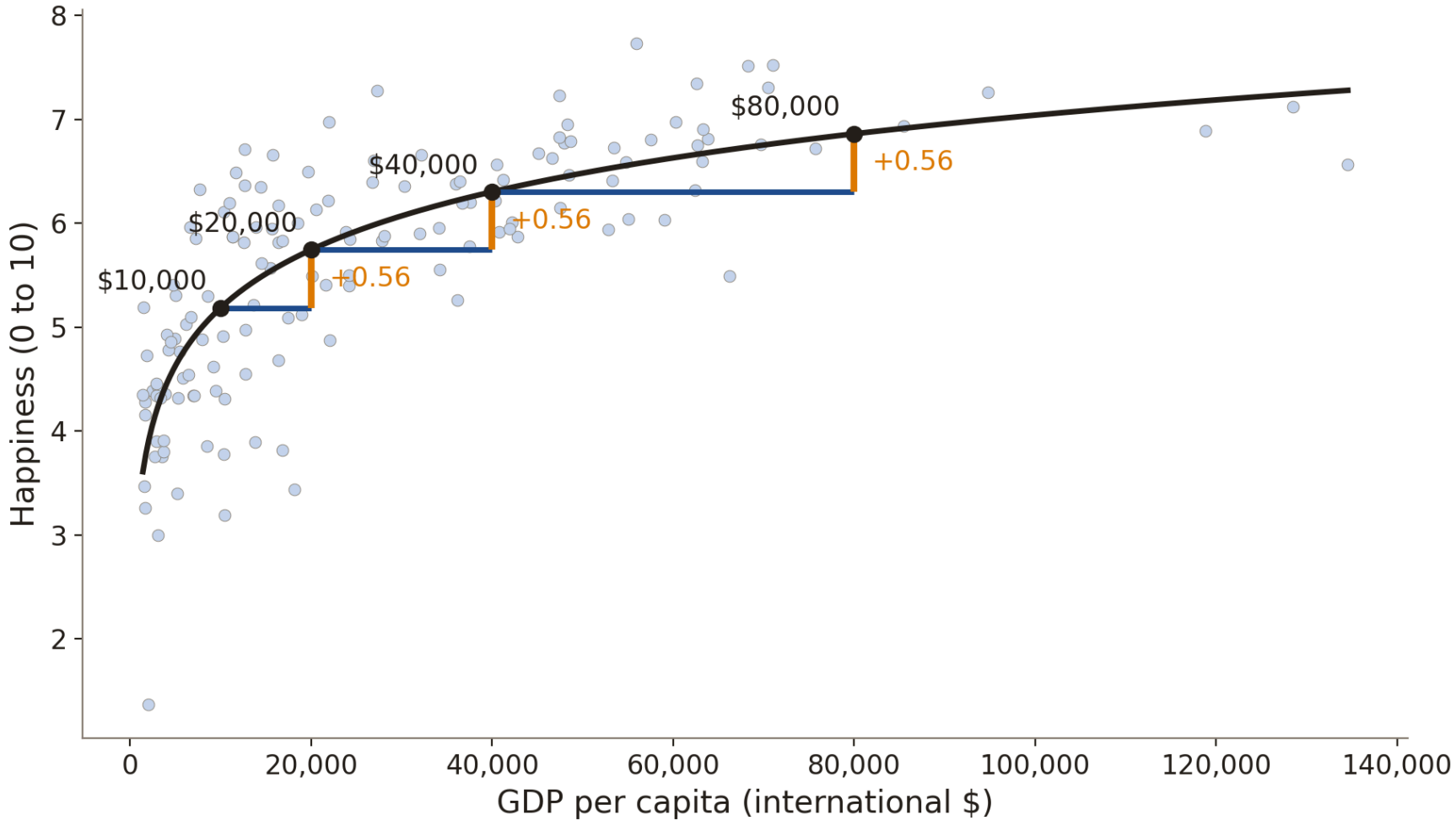
Logging GDP improves the functional form

$$\widehat{\text{happiness}} = -2.23 + 0.81 \times \log(\text{GDP per capita})$$



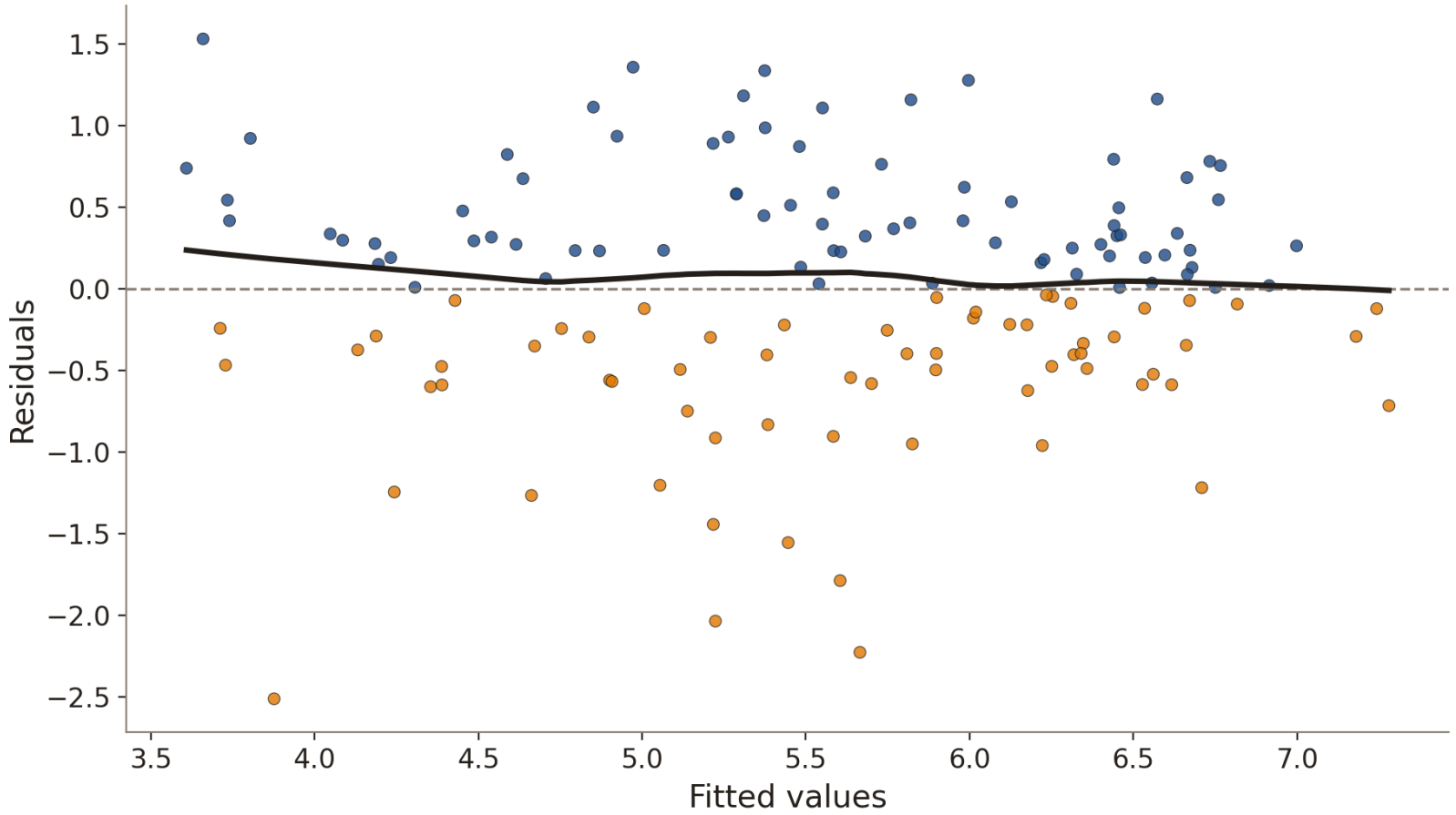
Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

Each doubling of GDP: same fitted difference



Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

With log GDP, no pattern in the residuals



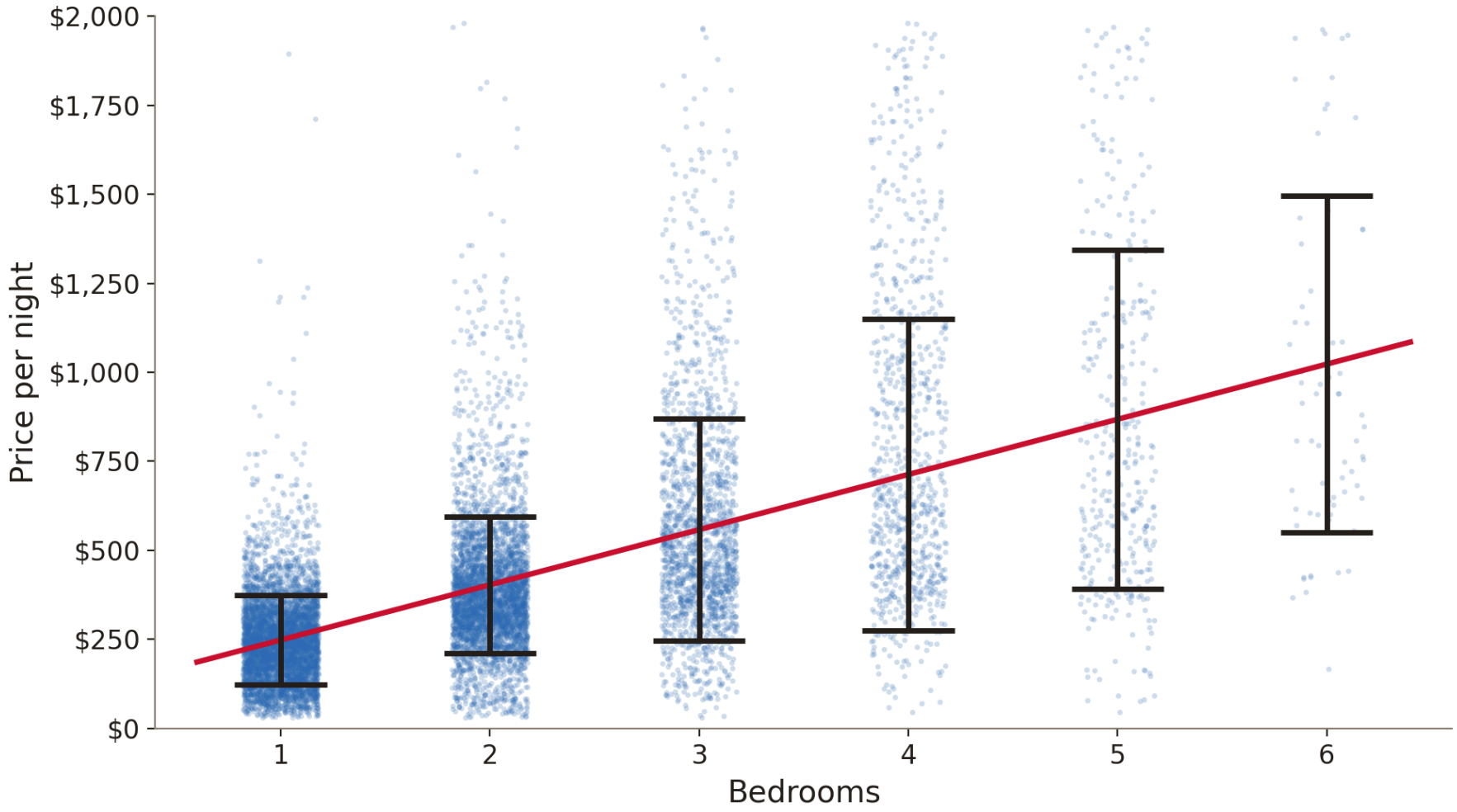
RESET: F = 0.76, p = 0.47

Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

Heteroskedasticity in Airbnb

Residual variance is larger for bigger listings

$$\widehat{\text{price}} = 92.89 + 155.01 \times \text{bedrooms}$$



Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings; the 99.8% with up to 6 bedrooms are shown

Breusch–Pagan: regress squared residuals on X

```
from statsmodels.stats.diagnostic import het_breuschpagan

model = ols("price ~ bedrooms", data=airbnb).fit()
bp = het_breuschpagan(model.resid, model.model.exog)
print("LM statistic:", round(bp[0], 1))
print("p-value:", round(bp[1], 4))
```

```
LM statistic: 1769.0
p-value: 0.0
```

p is below 0.1, so we reject the null of homoskedasticity

Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings

Is OLS biased? Are usual standard errors valid?

OLS is still unbiased: unbiasedness needs the error to have a mean of 0 given X, and heteroskedasticity is about its variance

The usual standard error formula assumes the same error variance for every observation

With heteroskedasticity, the usual standard errors are biased, so the t-statistics, p-values, and confidence intervals based on them are invalid

Robust standard errors: 1 change to `fit()`

```
model = ols("price ~ bedrooms", data=airbnb).fit()
robust = ols("price ~ bedrooms", data=airbnb).fit(cov_type="HC3")
```

	Usual	Robust
Coefficient on bedrooms	155.01	155.01
Standard error	1.85	3.66
95% confidence interval	151.4 to 158.6	147.8 to 162.2

Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings

Robust standard errors are a safe default

If the errors are homoskedastic, then usual and robust standard errors are both valid (robust ones in large samples)

If the errors are heteroskedastic, then only robust standard errors are valid, and only in large samples

Always plot residuals against fitted values

A curved pattern: try a log or a square of X, then run RESET again

Residual variance that differs across fitted values: report robust standard errors

No pattern: no evidence of either problem

Next week: measurement error